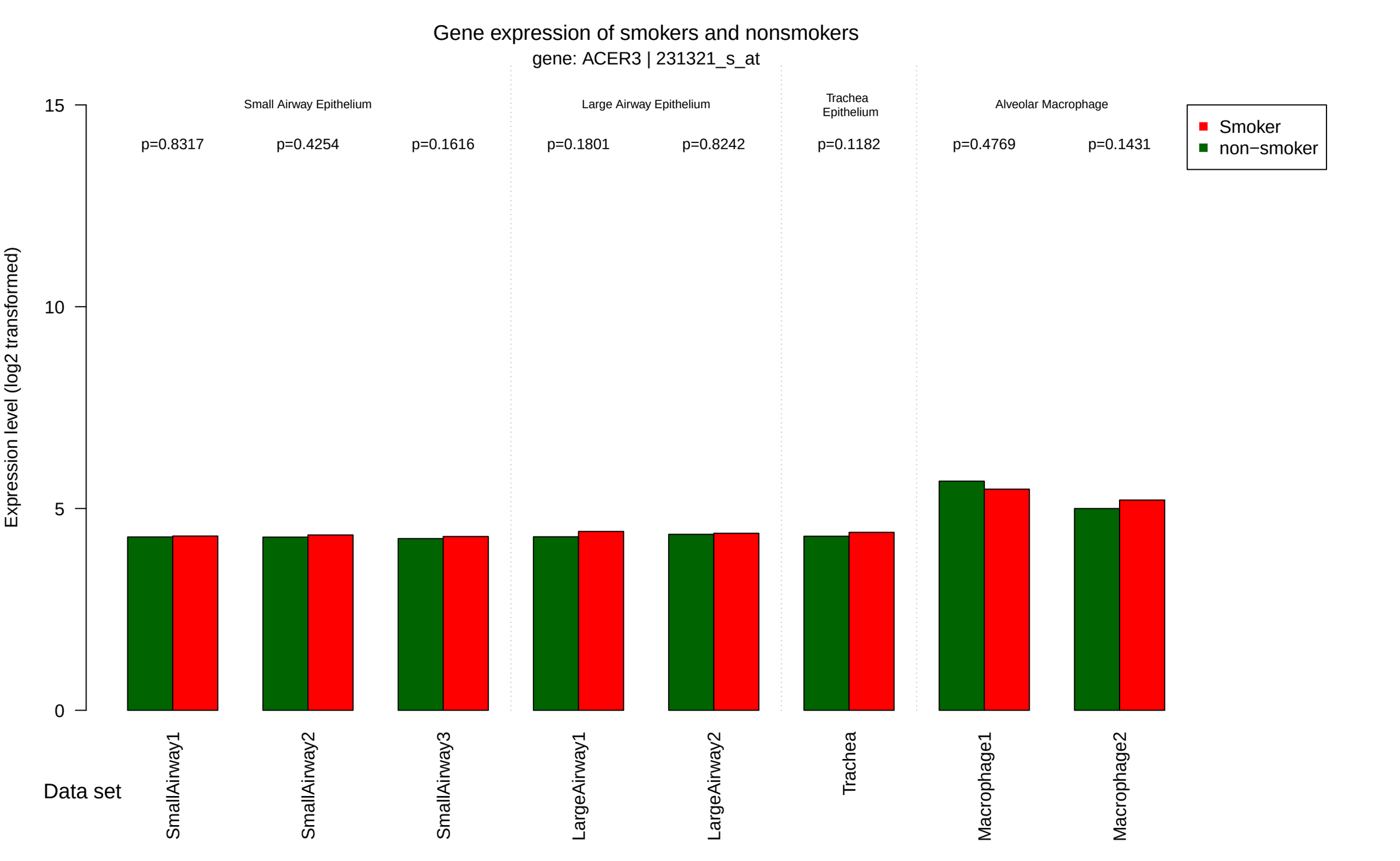
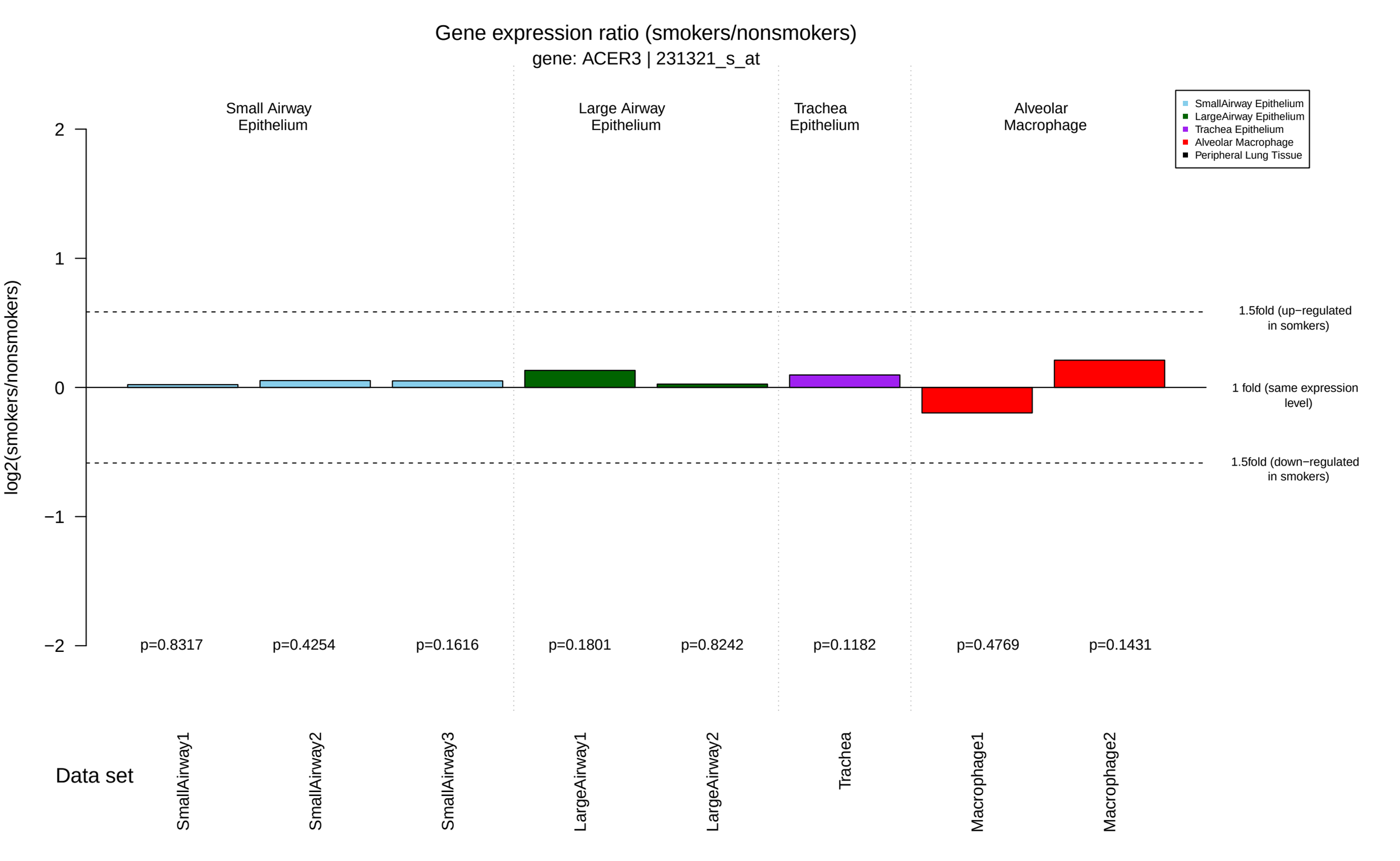
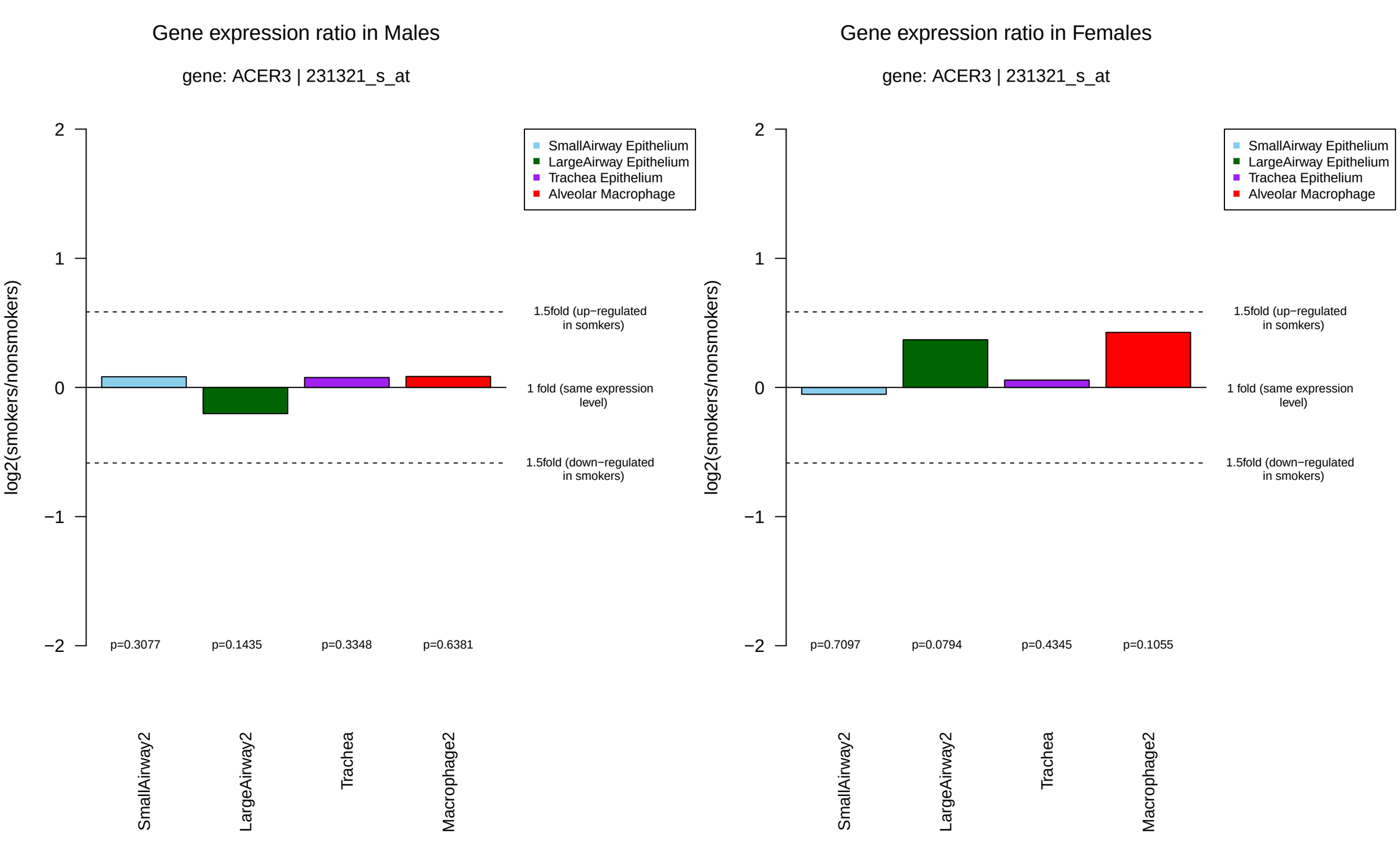
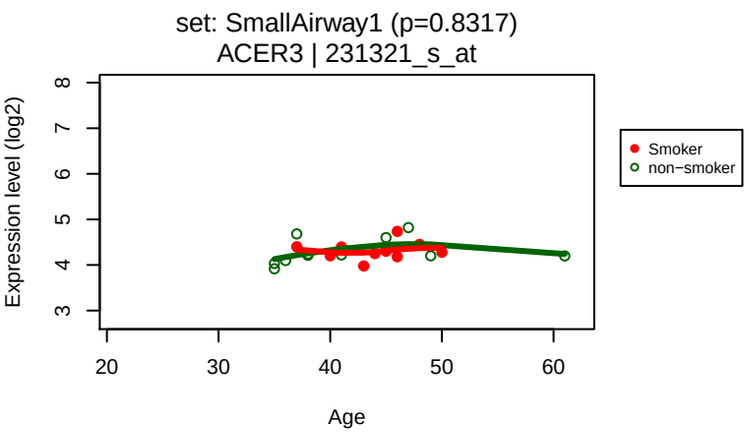
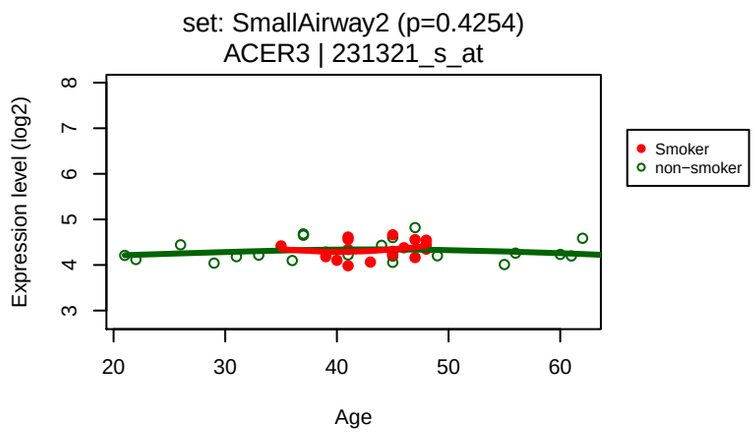
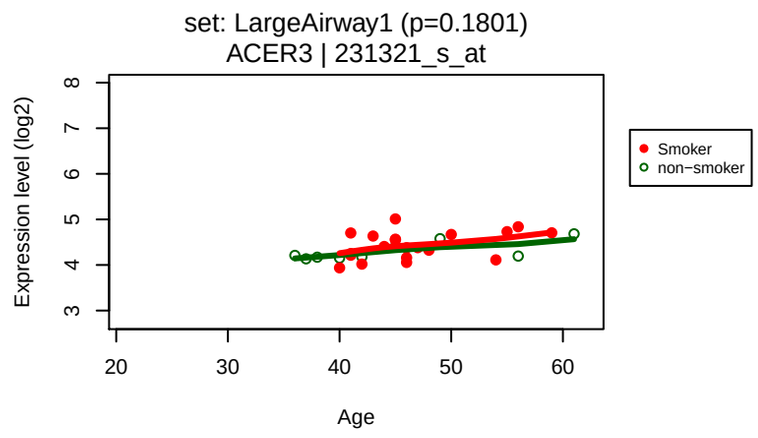

Gene expression of smokers and nonsmokers
gene: ACER3 | 231321_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8317
p=0.4254
p=0.1616
p=0.1801
p=0.8242
p=0.1182
p=0.4769
p=0.1431
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

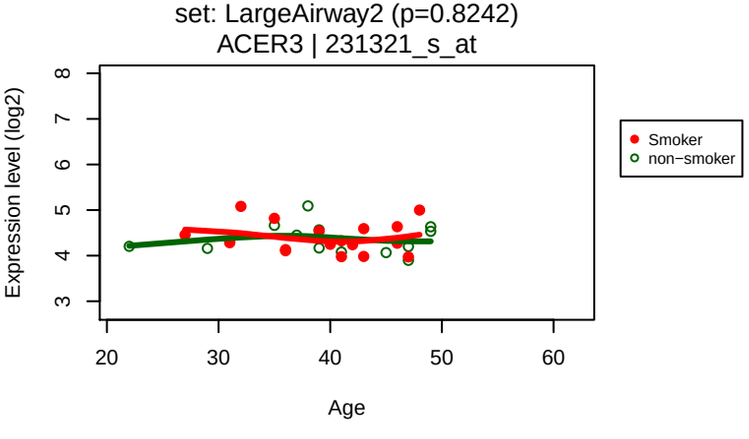

Gene expression ratio (smokers/nonsmokers)
gene: ACER3 | 231321_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8317
p=0.4254
p=0.1616
p=0.1801
p=0.8242
p=0.1182
p=0.4769
p=0.1431
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

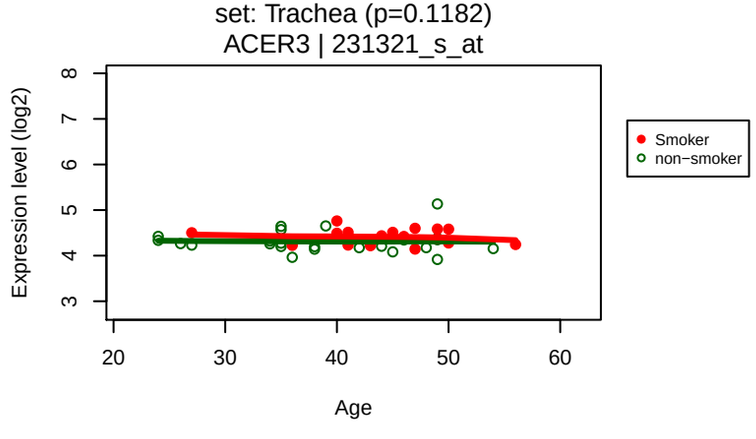

Gene expression ratio in Males
gene: ACER3 | 231321_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3077
p=0.1435
p=0.3348
p=0.6381
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACER3 | 231321_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7097
p=0.0794
p=0.4345
p=0.1055
SmallAirway2
LargeAirway2
Trachea
Macrophage2

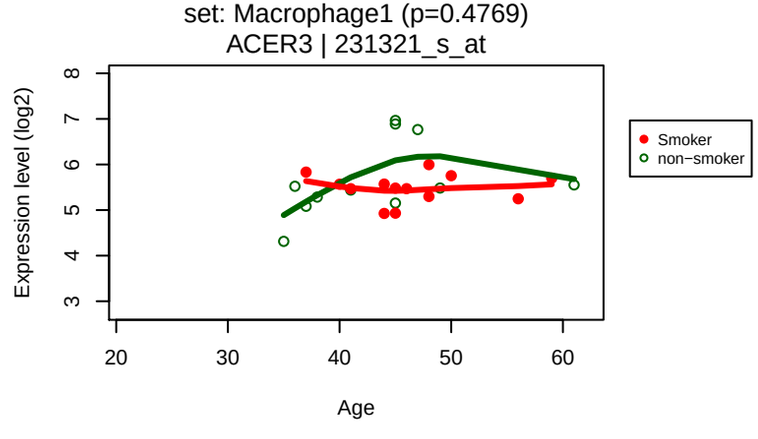

set: SmallAirway1 (p=0.8317)
ACER3 | 231321_s_at
Expression level (log2)
Age
Smoker
non−smoker

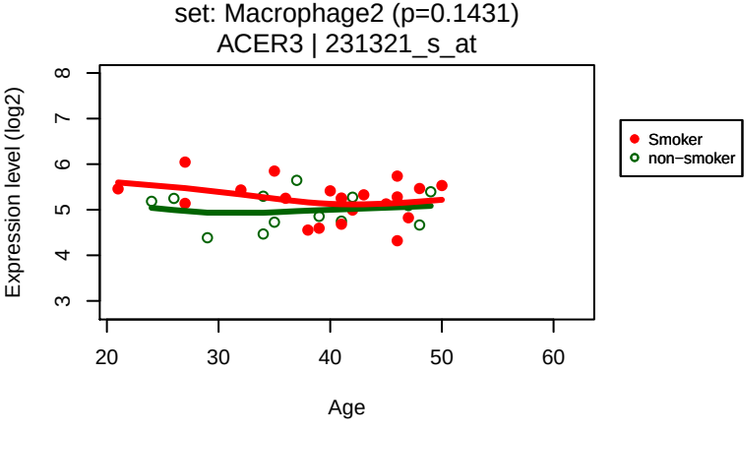

set: SmallAirway2 (p=0.4254)
ACER3 | 231321_s_at
Expression level (log2)
Age
Smoker
non−smoker

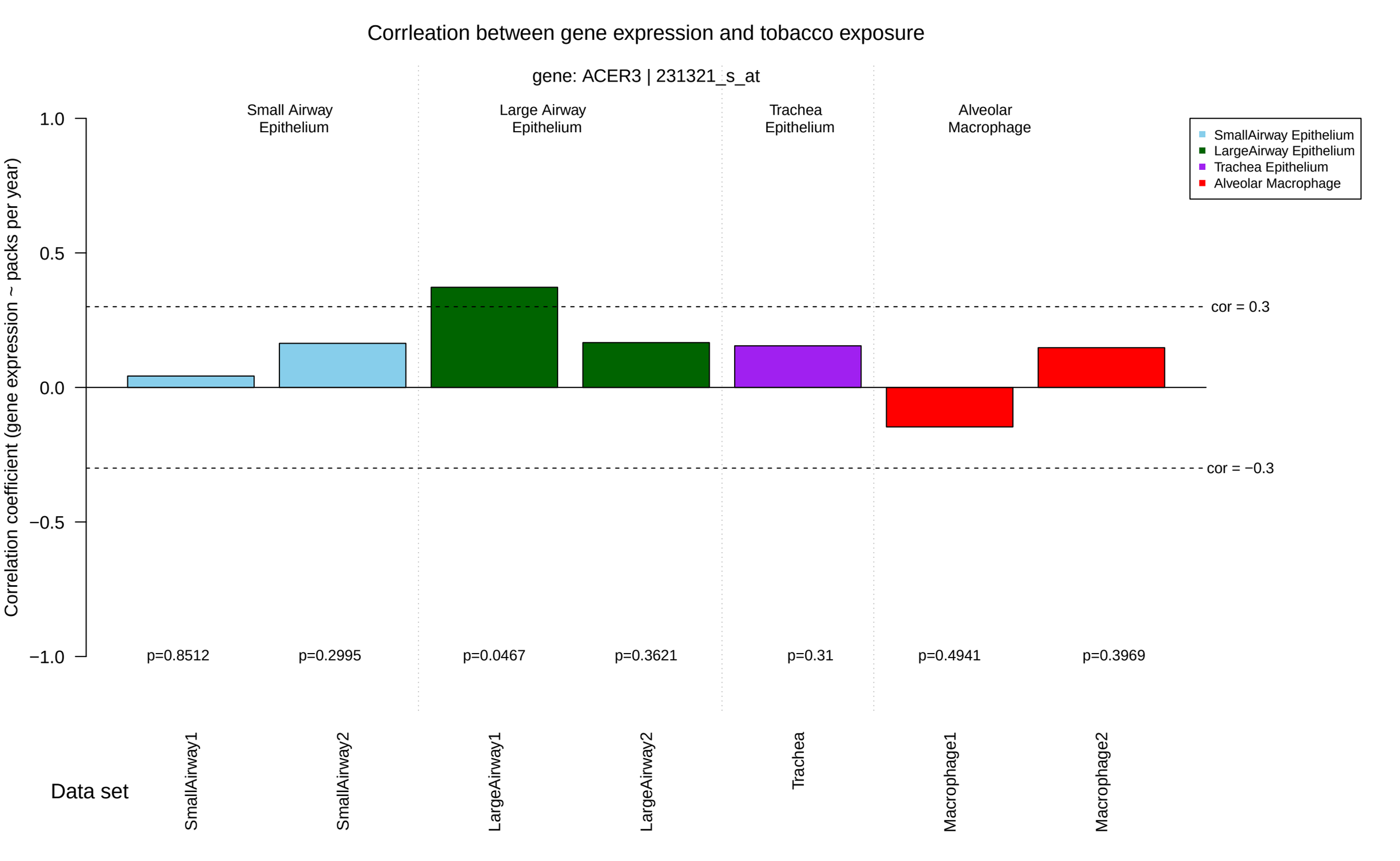

set: LargeAirway1 (p=0.1801)
ACER3 | 231321_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8242)
ACER3 | 231321_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1182)
ACER3 | 231321_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4769)
ACER3 | 231321_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1431)
ACER3 | 231321_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACER3 | 231321_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8512
p=0.2995
p=0.0467
p=0.3621
p=0.31
p=0.4941
p=0.3969
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2